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PM-JAY AWAAJSAHAYAK - TECH STACK SHEET

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Project: PM-JAY AwaajSahayak (Voice-First Healthcare Access Platform)

Problem Statement: ID - 3 (NIRMAN LAB-SEC2-73 / SH26)

Architecture: Zero-Data Missed Call Telephony & Assisted Kiosk Platform

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1. FRONTEND TECHNOLOGIES

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* HTML5

- Purpose: Provides the core semantic structure for the web platform,
including the tactile feature-phone container, the ASHA assistance desk,
and the survey data tables.

* CSS3 (Custom Variables & Modern Layouts)

- Purpose: Implements the visual styling, tactile Nokia-style feature phone
chassis, glowing LED screen effects, responsive grids, and accessible
high-contrast color schemes tailored for rural/elderly accessibility.

* JavaScript (ES6+)

- Purpose: Drives all client-side interactive logic, phone state machine
(IDLE -> DIALING -> RINGING -> ACTIVE_CALL), keypad event handling,
and dynamic beneficiary data rendering.

* Google Web Fonts (Plus Jakarta Sans & Noto Sans Devanagari)

- Purpose: Ensures clean, legible Latin typography for administrative tasks
and authentic Devanagari script rendering for Hindi and Marathi text.

2. AUDIO & SPEECH SYNTHESIS ENGINE

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* Web Speech API (SpeechSynthesis & SpeechSynthesisUtterance)

- Purpose: Serves as the zero-cost voice engine that speaks dynamic healthcare entitlement guidance aloud in regional languages without external paid APIs.

* Custom Devanagari Phonetic Voice Resolver (JavaScript)

- Purpose: Solves the common browser issue where Marathi voice packs are missing by routing Devanagari Marathi text through available Indian Hindi TTS engines, ensuring speech audio never fails silently.

* Web Audio API (AudioContext, OscillatorNode, GainNode)

- Purpose: Synthesizes realistic DTMF (Dual-Tone Multi-Frequency) keypad press tones and telephone ringing sound frequencies entirely in code.

3. BACKEND & TELEPHONY SIMULATION

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* Python 3.11 (http.server & socketserver)

- Purpose: Acts as a lightweight, zero-dependency local web server to host the application, serve static assets, and handle CSV/PDF file downloads.

* Telecom Missed-Call & IVR Architecture Model (Twilio / Exotel / Netcore)

- Purpose: Architectural design for production telephony: captures Caller Line Identification (CLI) on a toll-free number, disconnects to avoid cost, and schedules an automated outbound voice callback.

4. DATA MANAGEMENT & STORAGE

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* In-Memory / JSON Data Structure

- Purpose: Mocks the National Health Authority (NHA) database, storing beneficiary profiles, ABHA IDs, ₹5 Lakh coverage balances, and district hospital registries.

* CSV (Comma-Separated Values)

- Purpose: Stores the primary ground survey dataset (survey_data.csv) containing real household responses on PM-JAY awareness, literacy, and voice-tool readiness.

5. REPORT & DOCUMENTATION GENERATION

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* ReportLab 5.0 (Python PDF Library)

- Purpose: Programmatically compiles and renders the formal, publication-quality 6-page academic project report (PM_JAY_AwaajSahayak_Project_Report.pdf) with custom two-pass dynamic page numbering ("Page X of 6").